

Determining the Crystalline Structure of Metals through X-Ray Diffraction

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Abstract

This experiment set out to measure X-ray diffraction patterns of various solid samples in order to describe their crystal structure at the atomic/molecular level. An already calibrated Siemens !!!!!MODEL NAME!!!! X-Ray diffractometer was used to send X-rays into Copper (Cu), Nickel (Ni), CuNi alloy, Titanium (Sn), Lead (Pb), PbSn alloy and two mystery samples named A and B.

Any crystalline solid consists of a regular 3-dimensional packing of cells, the shape of which corresponds to one of fourteen Bravais lattices. These 3D patterns create multiple parallel planar layers within the crystal of various orientations. X-Rays travelling through these planes will diffract off the atoms and constructively interfere at the detector, creating a peak in photon count in the observed diffraction pattern. From that peak, Bragg's law $n\lambda = 2d \sin \theta$ (where $n = 1$ for a first-order plane) can be used to deduce the inter-planar distance d of the diffraction plane of the crystal. Using this inter-planar distance and the corresponding Miller indices, the Bravais lattice can be determined and the lattice parameter(s) calculated. The Cu and Ni pure samples and alloys were found to have a Face-Centered Cubic (FCC) Bravais lattice, with single lattice parameters [...] respectively. The alloy samples of these two were found to have [...]. The Sn and Pb samples were found to have a lattice of [...] and parameters [...] respectively, and their alloy [...], [...]. Finally, the mystery samples A and B were determined to be [...] and [...], with lattices [...] and parameters [...], respectively.

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1 Introduction

A solid can serve as a diffraction grating for X-rays of wavelengths similar lengths to the the inter-atomic distances. Similarly to how light diffracts, one can obtain a relation between the inter-planar distance d , the incident angle θ , and the wavelength λ :

$$n\lambda = 2d \sin(\theta) \quad (1)$$

with $n = 1$ for our purposes. Equation 1 is formally called Bragg's Law. Its utility becomes apparent when one ponders the crystalline structure of solids. Suppose the individual atoms of a crystal could be observed. While varying the angle the observer makes with the plane of the solid, there will be a moment where the atoms appear to line up. Given that a crystal is 3-dimension, the atoms will be aligned in parallel planes. Not all observed planes are equal however. Figure ?? shows a plane formed from a cubic arrangement of atoms. Notice that a plane could also have been formed from a face of the cube (in the y-z plane for example), or by aligning four diagonal atoms (the plane could be cutting through the y-axis and x-axis, for example). These planes are differentiated by their 'Miller index'.

As briefly mentionned above, the atoms in Figure ?? are arranged as a cube. This set of arranged atoms is a called a cell, and they can borrow their symmetries from one of fourteen Bravais lattices, each grouped into one of seven crystal systems. Figure ?? shows these lattices and their groups.

If an X-ray is sent to a solid with an angle θ such that (1) is satisfied, then constructive interference occurs between the different rays, and a peak in the intensity may be observed at a detector placed 2θ with respect to the incident X-rays. From these peaks, the inter-planar distances can be obtained. This inter-planar distance is related to the Bravais lattice and Miller indices by various equations. For a cubic system, this relation takes the form of

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2} \quad (2)$$

where h, k, l are the Miller Indices and a is the length of the cube edge. Similar equations for different crystal systems are given in [Appendix A](#).

2 Methods

2.1 Apparatus

The apparatus consisted of a pre-calibrated Siemens diffractometer was used to produce X-Rays, send them at the samples, and detect the diffractions. The diffractometer consisted primarily of an X-ray tube, a sample mounting plate, and a photon detector. The X-ray tube used Copper K_α radiation and was cooled using water. Two sets of filters of 0.6 mm were present on the X-ray tube and the detector, to prevent stray rays from leaving the tube and entering the detector. During data acquisition, the diffractometer was set to 40 kV and 40 mA. The mounting plate kept the sample in place with a spring-loaded mechanism. The samples were centered by aligning the middle of the sample with a red line marked on the mounting plate. The diffractometer changes the θ by rotating the mounting plate by some $\Delta\theta$ and moving the detector by $2\Delta\theta$.

A computer was connected to the diffractometer, with which a .dql measuring parameter file was given to the diffractometer and data was obtained from the detector as .RAW files. The measuring parameters were an initial angle of $2\theta_0 = 5.00^\circ$, angle increments of $\Delta\theta = 0.05^\circ$, with 2.00 s at each step, with a final angle of $2\theta_f = 110.00^\circ$. The total measuring time is roughly 1 h 05 m using the aforementioned parameters.

2.2 Data Acquisition and Analysis

Data acquisition was simple: Mount the desired sample onto the mounting plate, centering it with the red line. Then, on the computer, open the software program, select the desired parameter file, name the trial, and execute the job. After the job finishes, convert the .RAW file to .UXD. Finally, the .UXD files were transferred from the lab computer onto the experimentalists's for the data analysis.

3 Results

Three trials were done for Ni and Cu samples, though the data for the first Cu trial did not save correctly. For each of the alloys (including different percentage of mixing) one trial was done. In figure 2, the residuals for the Ni samples are shown, and from their numerical results in table ?? we can observe that random error from trial to trial is minimal compared with noise error provided by the Gaussian fits. The fitted gaussian sums with residuals for the Cu and Ni samples and alloys are shown in the same appendix, in figure 3. From these, the inter-planar distances were calculated and are shown in table 1. These were plotted against expected results for all 3 types of cubic Bravais lattices, and the best fit was found to be the Face-Centered Cubic (FCC) lattice, as shown below in figure 1. The lattice parameters were then calculated and are shown in table 1.

Sample	a_{exp} (Å)	a_{theory} (Å)	Δa (Å)
Ni	3.5223 ± 0.0001	3.5238	-0.0015 ± 0.0001
Cu	3.6126 ± 0.0001	3.6149	-0.0023 ± 0.0001
Cu ₂₅ Ni ₇₅	3.5415 ± 0.0002	3.5467	-0.0052 ± 0.0002
Cu ₅₀ Ni ₅₀	3.5629 ± 0.0003	3.5696	-0.0067 ± 0.0003
Cu ₇₅ Ni ₂₅	3.5864 ± 0.0002	3.5924	-0.0060 ± 0.0002

Table 1: Experimental and theoretical lattice parameters for Cu, Ni pure samples and alloys. The theoretical values for the alloys are calculated using Vegard’s law .

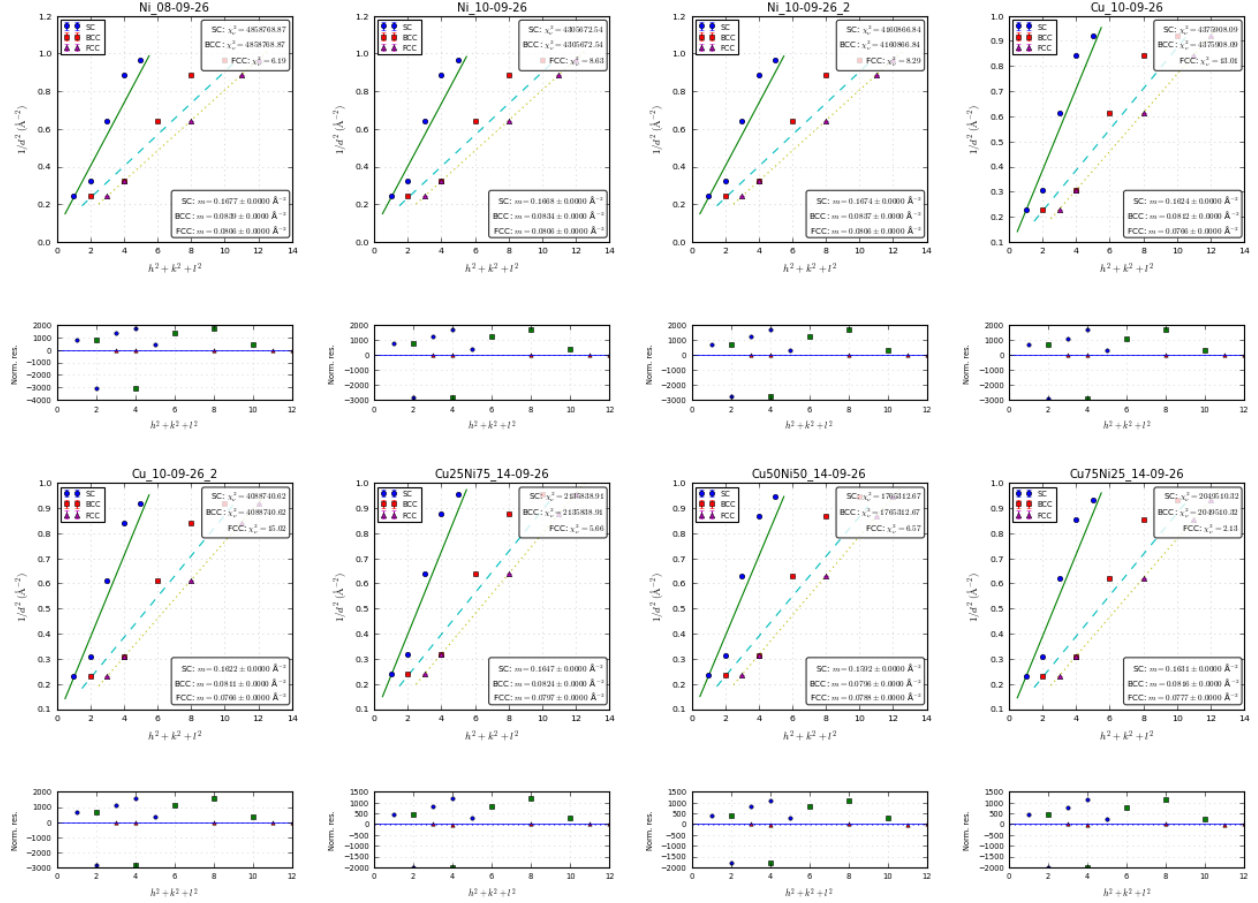


Figure 1: Determination of Cubic crystal structure and lattice parameter for Cu and Ni.

4 Discussion

The results for the Cu and Ni pure samples and alloys are consistent with the literature values, with the experimental values being within 0.2% of the theoretical values. The small discrepancies can be attributed to background noise in the acquired data. The determination of the Face-Centered Cubic (FCC) Bravais lattice for these samples is also consistent with known crystal structures for Cu and Ni. While running multiple trials does not reduce error much, it could still be worth it, as random error is of similar order as the noise error. The results for the Pb, Sn, and PbSn alloys are still to be determined, as well as the mystery samples A and B.

PLAN

Accomplished

- Take data for all samples (pure Cu, Ni, Pb, Sn, CuNi alloys, PbSn alloys, and mystery samples A and B).
- Fit data using gaussian sum to find peak positions and uncertainties.
- Calculate inter-planar distances and uncertainties.
- Match inter-planar distances to Miller indices and cubic Bravais lattices for Cu, Ni, and CuNi alloys.
- Calculate lattice parameters and uncertainties for Cu, Ni, and CuNi alloys.

To Do

- Do more runs for samples with only 1 run
- Produce gaussian sum fits for Pb, Sn, and PbSn alloys.
- Calculate inter-planar distances and uncertainties for Pb, Sn, and PbSn alloys.
- Determine how to match inter-planar distances to Miller indices for non-cubic Bravais lattices (Pb, Sn, and PbSn alloys).
- Determine Bravais lattices for Pb, Sn, and PbSn alloys.
- Calculate lattice parameters and uncertainties for Pb, Sn, and PbSn alloys.
- Buildin on the method developed above, determine the Bravais lattices and lattice parameters for mystery samples A and B.

References

Appendix A. Inter-planar distance, Bravais Lattices and Miller indices

For the following equations: d is the inter planar distance, a, b, c the inter-atomic distances, h, k, l the indices for a given Miller index (hkl).

Cubic:

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

Tetragonal:

$$\frac{1}{d^2} = \frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2}$$

Hexagonal:

$$\frac{1}{d^2} = \frac{4}{3} \left(\frac{h^2 + k^2 + hk}{a^2} \right) + \frac{l^2}{c^2}$$

Orthorhombic:

$$\frac{1}{d^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

Vegard's law:

$$a_{\text{alloy}} = xa_{Cu} + (1 - x)a_{Ni}$$

Appendix B. Supplementary Figures and Tables

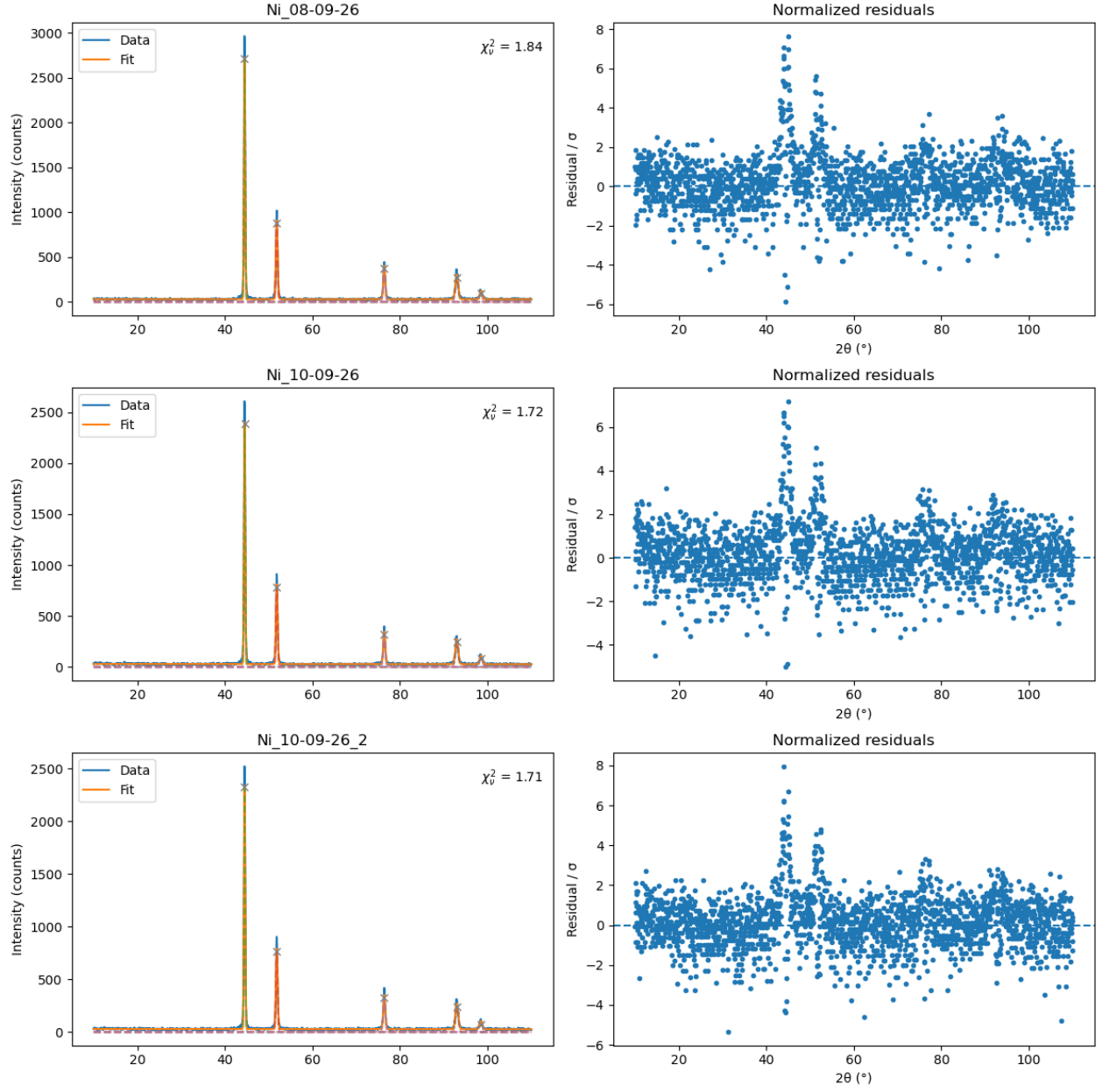


Figure 2: Gaussian sum fits of all pure Ni trials with residuals

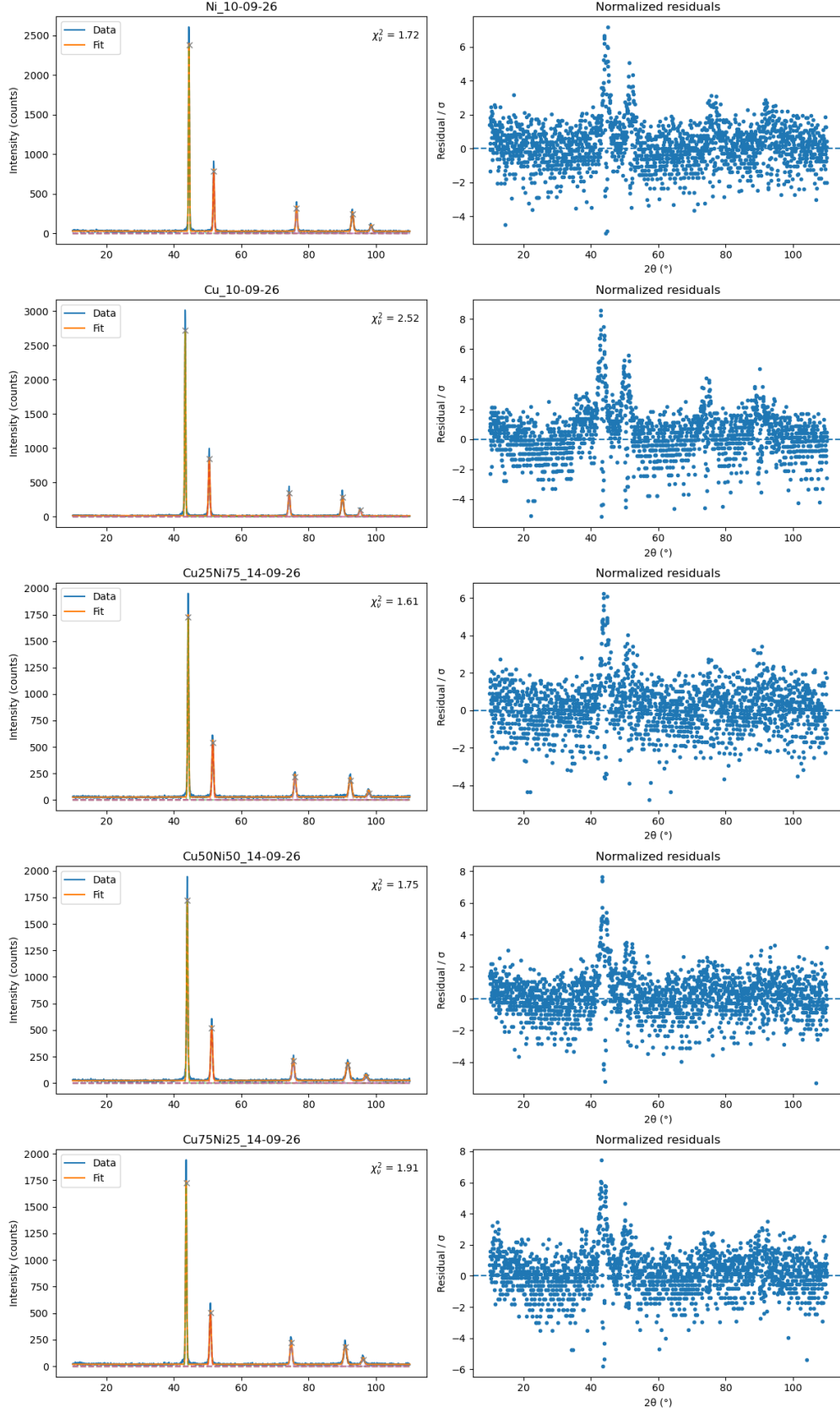


Figure 3: Gaussian sum fits with residuals of the Cu and Ni pure and alloy samples

Ni trial 1			Ni trial 2		
Peak	2θ (°)	d (Å)	Peak	2θ (°)	d (Å)
1	44.4819 ± 0.0011	2.03512 ± 0.00005	1	44.4843 ± 0.0011	2.03502 ± 0.00005
2	51.8330 ± 0.0023	1.76246 ± 0.00007	2	51.8331 ± 0.0024	1.76245 ± 0.00008
3	76.3993 ± 0.0041	1.24563 ± 0.00006	3	76.3933 ± 0.0045	1.24571 ± 0.00006
4	92.9684 ± 0.0059	1.06221 ± 0.00005	4	92.9701 ± 0.0062	1.06220 ± 0.00005
5	98.4832 ± 0.0109	1.01694 ± 0.00008	5	98.5025 ± 0.0127	1.01679 ± 0.00010
Ni trial 3			Cu trial 1		
Peak	2θ (°)	d (Å)	Peak	2θ (°)	d (Å)
1	44.4810 ± 0.0012	2.03517 ± 0.00005	1	43.3661 ± 0.0011	2.08487 ± 0.00005
2	51.8307 ± 0.0024	1.76253 ± 0.00008	2	50.4888 ± 0.0024	1.80618 ± 0.00008
3	76.4023 ± 0.0045	1.24558 ± 0.00006	3	74.2095 ± 0.0043	1.27687 ± 0.00006
4	92.9767 ± 0.0062	1.06214 ± 0.00005	4	90.0228 ± 0.0054	1.08915 ± 0.00005
5	98.4665 ± 0.0135	1.01707 ± 0.00010	5	95.2553 ± 0.0097	1.04266 ± 0.00008
Cu trial 2			Cu ₂₅ Ni ₇₅		
Peak	2θ (°)	d (Å)	Peak	2θ (°)	d (Å)
1	43.3667 ± 0.0011	2.08484 ± 0.00005	1	44.2433 ± 0.0015	2.04554 ± 0.00006
2	50.4882 ± 0.0024	1.80620 ± 0.00008	2	51.5406 ± 0.0034	1.77176 ± 0.00011
3	74.2000 ± 0.0043	1.27701 ± 0.00006	3	75.9226 ± 0.0065	1.25226 ± 0.00009
4	90.0104 ± 0.0057	1.08927 ± 0.00005	4	92.3239 ± 0.0085	1.06793 ± 0.00008
5	95.2715 ± 0.0098	1.04253 ± 0.00008	5	97.7855 ± 0.0166	1.02232 ± 0.00013
Cu ₅₀ Ni ₅₀			Cu ₇₅ Ni ₂₅		
Peak	2θ (°)	d (Å)	Peak	2θ (°)	d (Å)
1	43.9763 ± 0.0016	2.05734 ± 0.00007	1	43.6499 ± 0.0015	2.07197 ± 0.00007
2	51.2219 ± 0.0036	1.78204 ± 0.00012	2	50.8455 ± 0.0035	1.79434 ± 0.00011
3	75.4112 ± 0.0071	1.25948 ± 0.00010	3	74.8037 ± 0.0062	1.26819 ± 0.00009
4	91.6044 ± 0.0098	1.07443 ± 0.00009	4	90.8289 ± 0.0083	1.08157 ± 0.00008
5	97.0016 ± 0.0200	1.02849 ± 0.00016	5	96.1327 ± 0.0160	1.03546 ± 0.00013

Table 2: Measured X-ray diffraction peak angular positions and corresponding interplanar spacings (d) for the Ni, Cu, and Cu–Ni alloy samples.